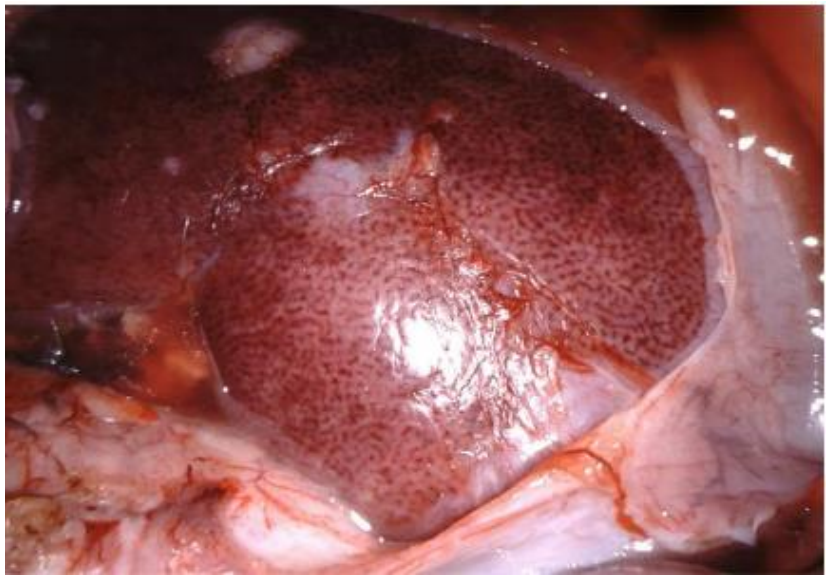
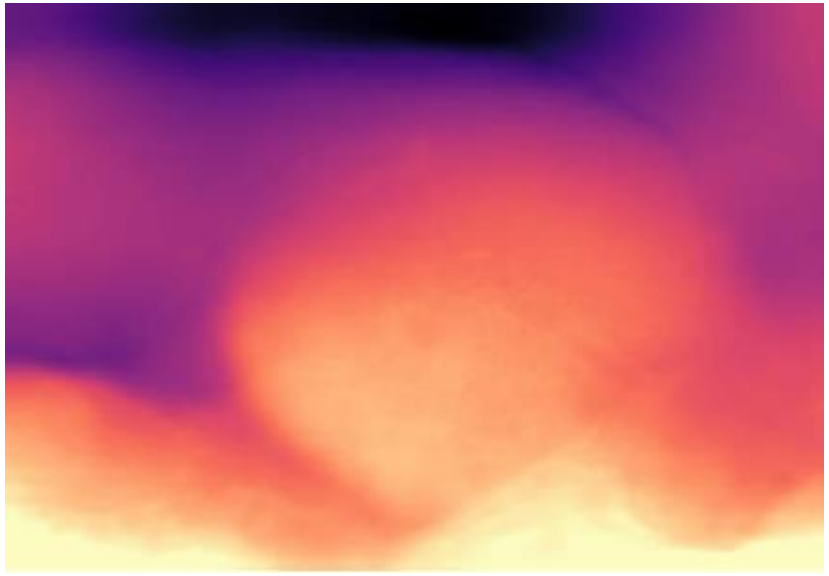
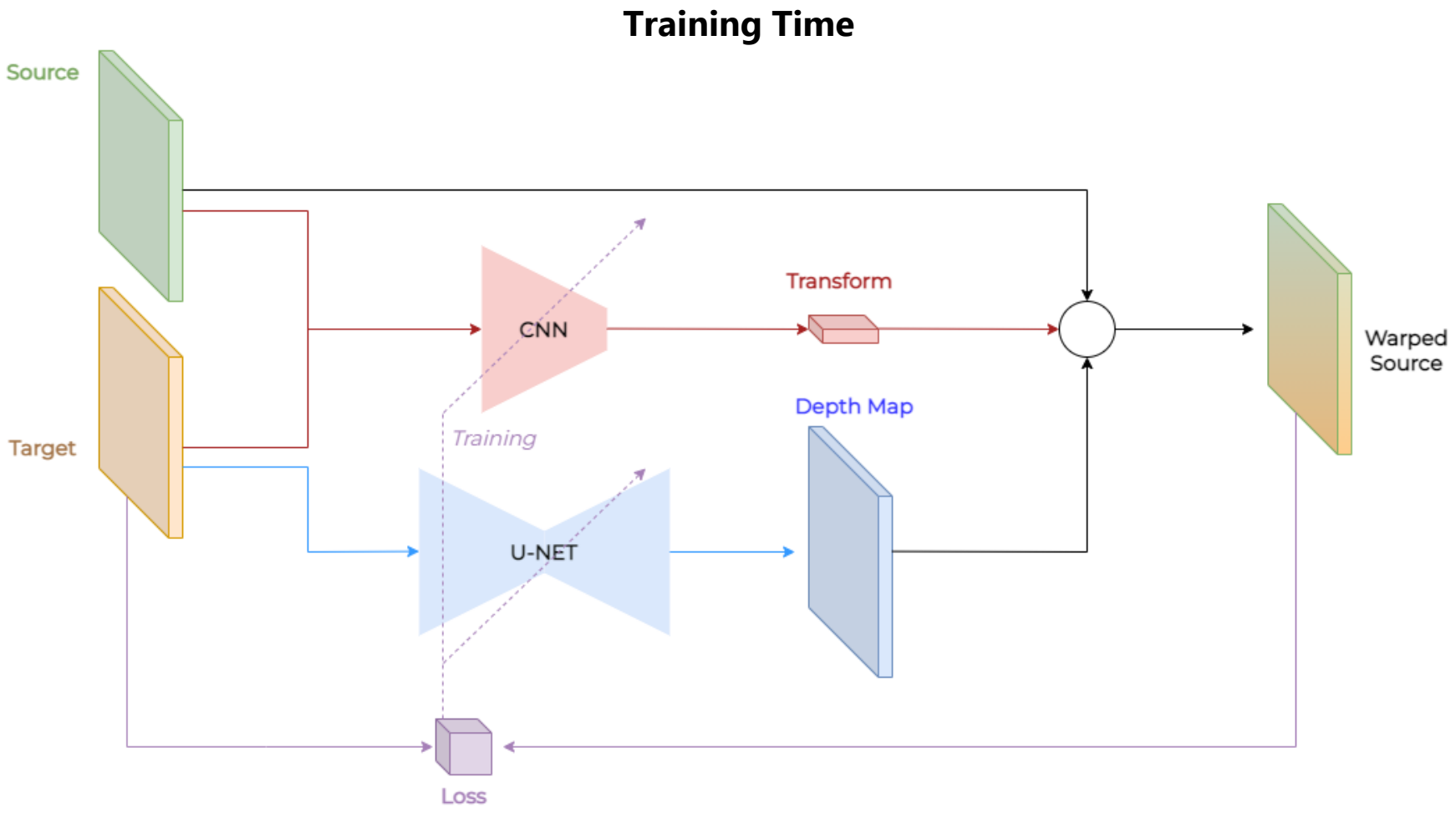
Single-Image Depth Estimation

Planning and Datasets

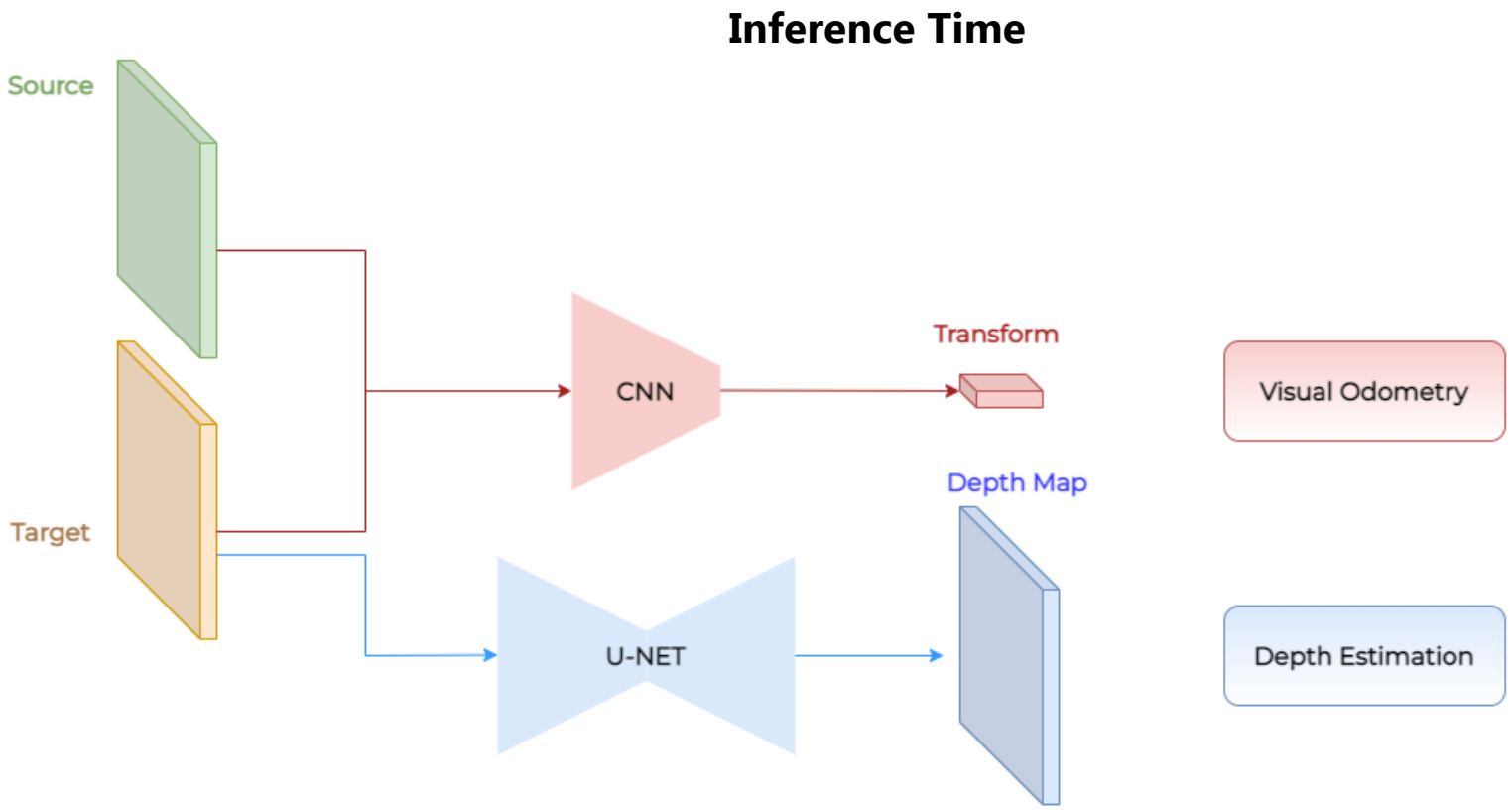




State of the Art Framework



State of the Art Framework



Dataset

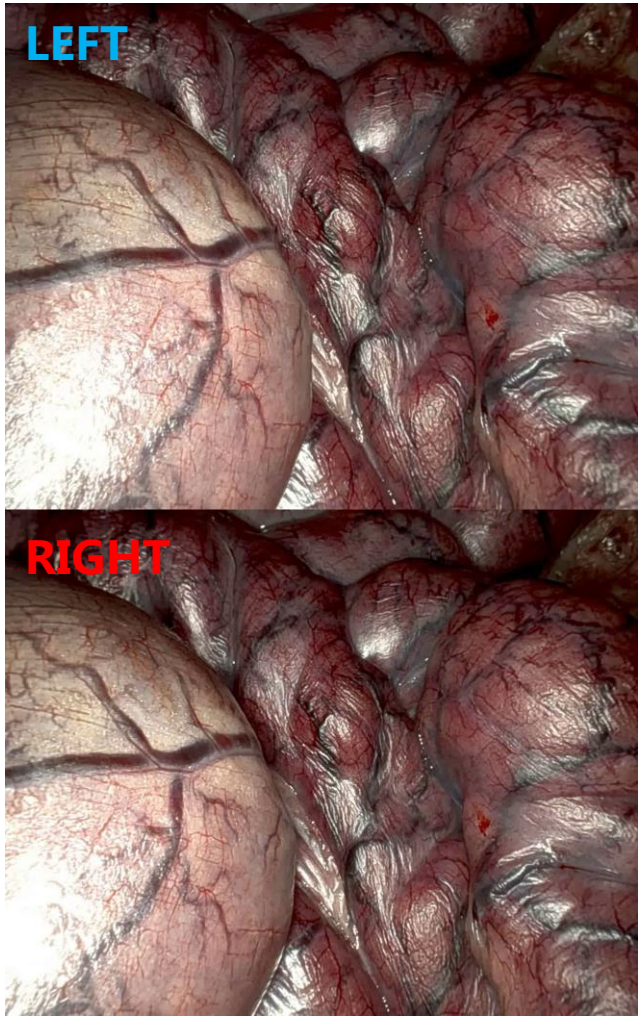
EndoVis / SCARED

- **55 Videos, 25 fps, ~20 sec, stereo**
- Camera pose @ each frame

→**28’000 frames**, available as:

- Left-right pairs
- Source-target pairs

(at training time)

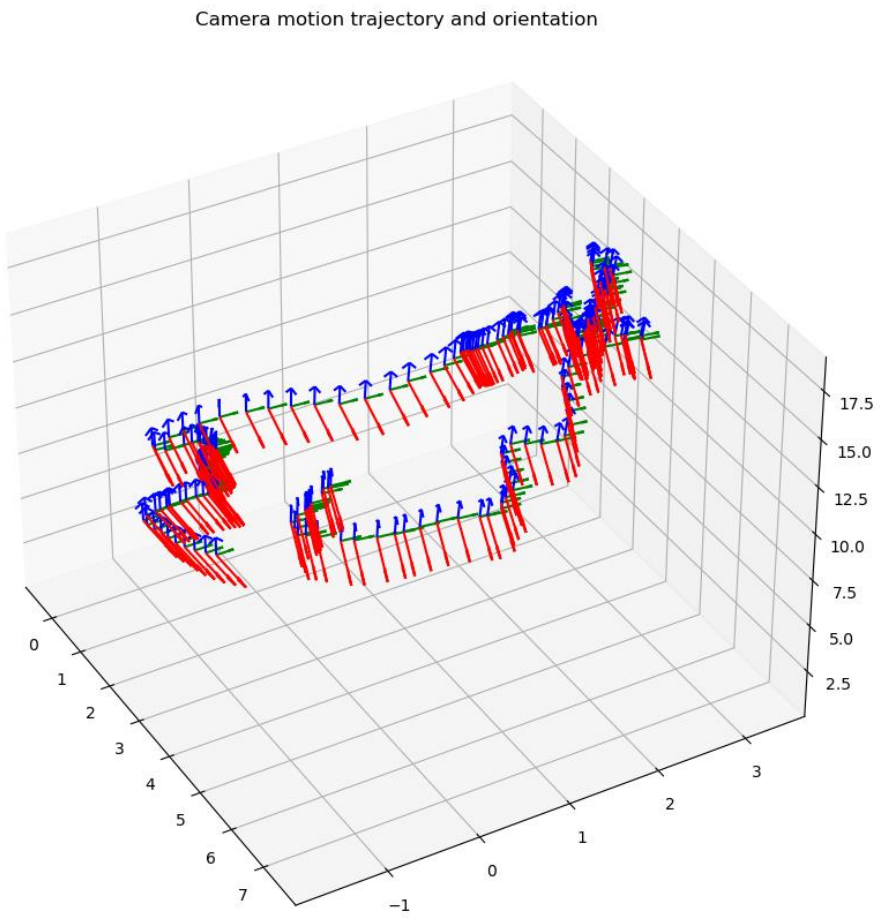


Dataset

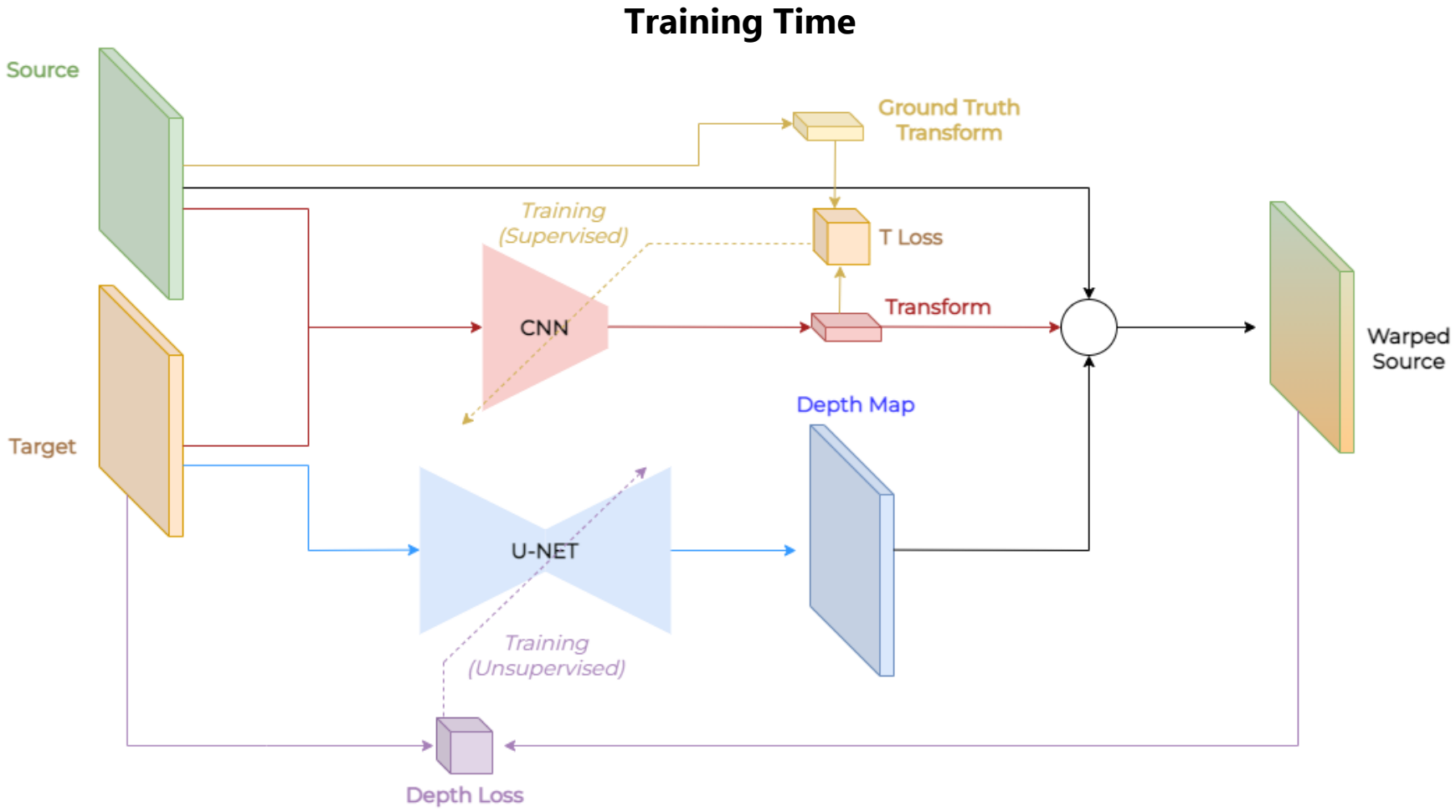
EndoVis / SCARED

- 55 Videos, 25 fps, ~20 sec, stereo
- **Camera pose @ each frame**

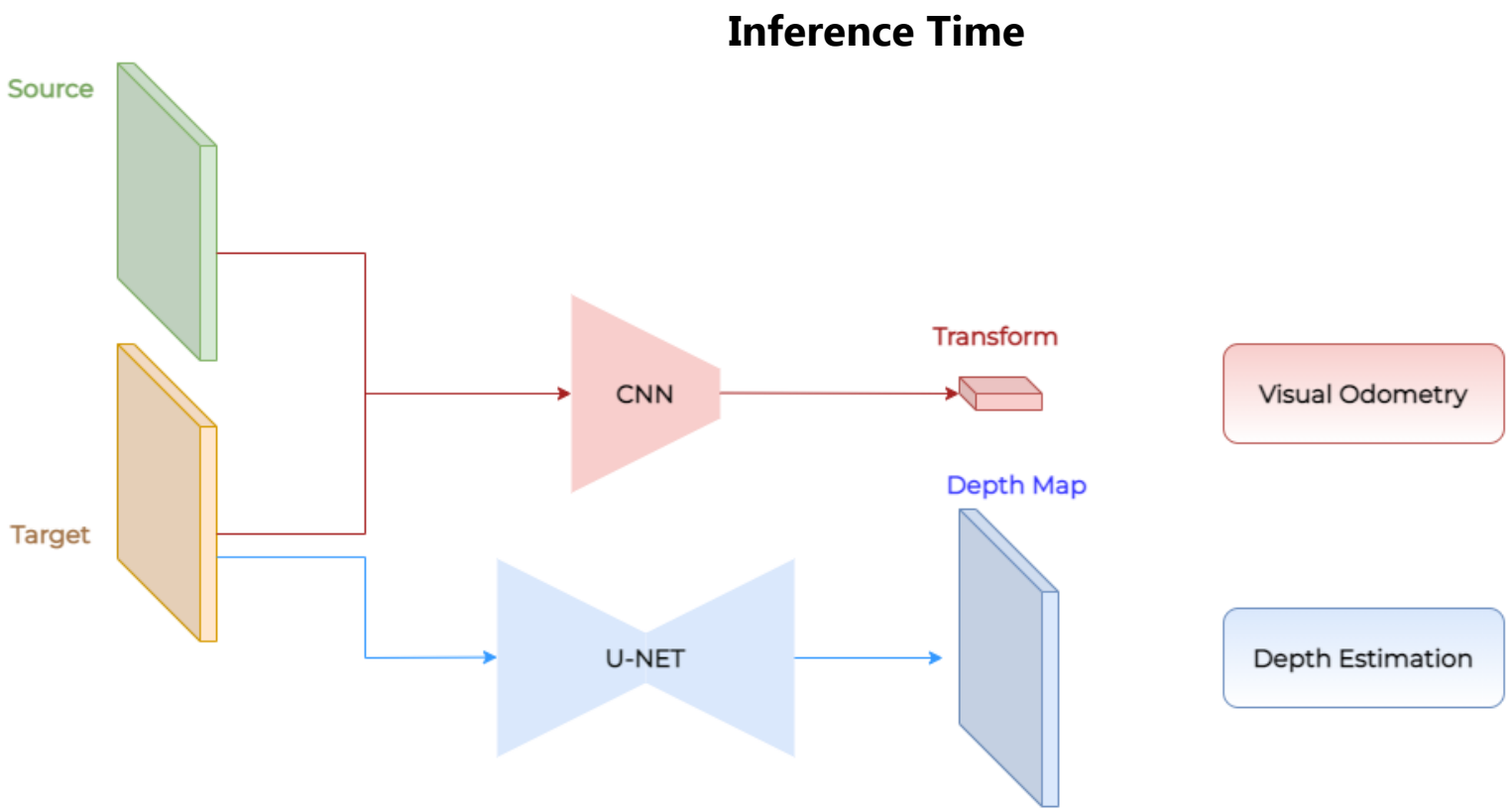
→Inter-frame transformation can be calculated among frame *k* and *k-1*, but also *k* and *k-2, etc*



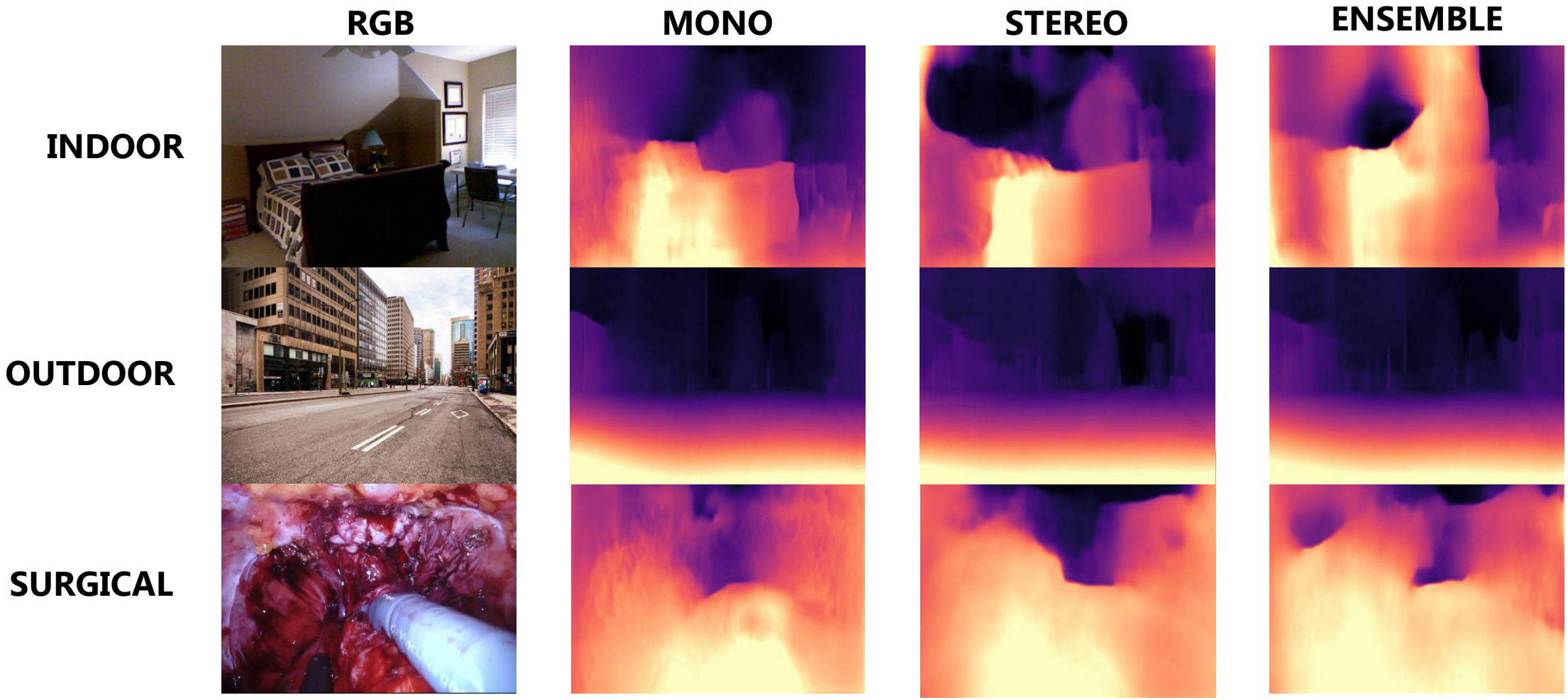
Our Framework



Our Framework



Monodepth2 – Validation tests



Progress - Technical

- State of the art Qualitative Validation (*monodepth2*)

- Started downloading the dataset (it's 260GB…)
- Dataset cleanup, conversion and Torch Pipeline Setups

- Data preprocessing – RGBs and Transforms
- Augmentations (?)